



Remote sensing and in situ observations of harmful algal blooms in dongzhang reservoir: implications for water quality management

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Abstract Harmful Algal Blooms (HABs) cause significant ecological damage and public health issues in freshwater reservoirs globally. While in situ sampling is essential for monitoring HABs, it is often costly and time-consuming. Remote sensing offers a rapid and cost-effective alternative for detecting HABs. This study integrates Sentinel-2 multispectral imagery with in-situ observations to monitor HABs in Dongzhang Reservoir, Fujian, China. We analyzed the Normalized Difference Vegetation Index and Normalized Difference Chlorophyll Index (NDCI) to monitor chlorophyll-a concentrations in this dynamic water body. In situ water samples were collected at three locations to determine the phytoplankton density, and

these counts were correlated with spectral indices. Our findings indicate that the Dongzhang Reservoir experiences high algal biomass during spring and summer, as evidenced by seasonal variations in spectral index values. The NDCI, which leverages the red edge wavelength, provided the most accurate identification of chlorophyll-a. Linear regression analysis revealed strong correlations between phytoplankton density and remote sensing values in spring, summer, and autumn of 2022, with R^2 values of 0.81, 0.74, and 0.62, respectively. Two key factors driving HAB proliferation in 2022 were total phosphorus levels and water surface temperatures, which were positively correlated with phytoplankton density ($R^2=0.83$ for total phosphorus and $R^2=0.81$ for water temperature). In 2023, total phosphorus remained the main factor, with an R^2 of 0.75. HAB events were more

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severe in 2022, affecting 34.41% of the reservoir area in May, compared with 22.15% in 2023. This study demonstrated that Sentinel-2 imagery captures greater spatial and temporal heterogeneity of algal blooms than traditional in situ samples do, highlighting the potential of remote sensing as a crucial tool for monitoring algal bloom dynamics in reservoirs and other water bodies. Complementing in situ samples with Sentinel-2 images, owing to their high temporal resolution, can increase monitoring efforts. The implementation of appropriate management strategies can mitigate HABs in Dongzhang Reservoir.

Keywords Phytoplankton · In situ monitoring · Remote sensing · Chlorophyll a · Nutrients

Introduction

Freshwater reservoirs play a critical role in supporting human needs such as irrigation, drinking water supply, and energy production (Carpenter et al., 2011; Lobo et al., 2021). However, these systems are increasingly threatened by harmful algal blooms (HABs), which pose risks to aquatic ecosystems, water quality, and public health. HABs release toxins that can cause respiratory and liver problems in humans, as well as livestock fatalities, while also contributing to hypoxic conditions, fish kills, and economic losses (Paerl et al., 2001; Davidson et al., 2016). Timely monitoring and early warning are vital for mitigating the impacts of HABs. Various approaches, including satellite and drone observations, biological and toxin analyses, molecular techniques, modeling, and citizen science, are used for HAB detection. However, traditional field sampling is often constrained by time, weather, and cost. In contrast, remote sensing (RS) provides wide coverage, rapid data access, and frequent monitoring, making it a valuable tool for water quality assessment (Kislik et al., 2022; Zahir et al., 2024a; 2024b; Kumar et al. (2025)). Nonetheless, RS is limited by cloud cover, atmospheric interference, and its focus on surface conditions, which may miss subsurface blooms. Optical complexity from suspended sediments and dissolved organic matter can further affect accuracy, while spatial resolution and revisit intervals may restrict detection of small or short-lived blooms. Thus, integrating RS with in-situ data remains

essential for reliable HAB monitoring (Raghul & Porchelvan, 2024; Wang & Qin, 2025).

To monitor HAB dynamics in reservoirs, the Sentinel-2 A/B Multi-Spectral Imager (MSI) has shown significant performance due to its high spatial resolution (10–60 m), temporal resolution (5 days), and wide spectral range (Ansper & Alikas, 2018). Sentinel-2 provides high-resolution multispectral imagery, including bands specific for chlorophyll-a detection, a pigment present in HABs (Caballero et al., 2020). Sentinel-2 sensors have been used in several studies to investigate algal blooms in coastal areas (Pahlevan et al., 2017) and large inland lakes (Bramich et al., 2021a, 2021b; Ha et al., 2017). Despite their primary focus on land use monitoring, these sensors have proven useful for HAB monitoring in small reservoirs, offering more frequent observations compared to Landsat's 16-day revisit interval (Caballero & Navarro, 2021; Caballero et al., 2020). Sentinel-2 imagery is captured regularly, enabling consistent monitoring of HAB dynamics. The red and near-infrared bands are commonly used for HAB detection, as they provide crucial data on chlorophyll-a concentrations (Kislik et al., 2022; Wang et al., 2020).

By analyzing the spectral reflectance of the water surface, it is possible to estimate chlorophyll-a concentration and identify areas with high algal biomass. The data derived from Sentinel-2 imagery helps map the extent and severity of HABs, assessing potential risks to both the environment and human health. Combining in situ measurements and remote sensing data allows for effective monitoring of HABs in small water reservoirs, enabling quick responses from water quality managers and public health agencies. Studying algal blooms in small lakes and reservoirs spatiotemporally can help identify potentially dangerous conditions (Anderson et al., 2012; Kislik et al., 2022; Rodríguez-López et al., 2023).

In this study, we utilized in situ data and Sentinel-2 data (spectral indices) for the Dongzhang Reservoir in Fujian Province, China, to assess the effectiveness of Sentinel-2 imagery in monitoring HABs. Our objectives included evaluating Sentinel-2's ability to detect chlorophyll-a and establishing baseline water quality data. This study is the first to analyze the temporal dynamics of algal blooms in the Dongzhang Reservoir using Sentinel-2 imagery in Fujian, China, as Sentinel-2 data have not been used for this purpose in this reservoir previously. A local study

utilizing Landsat-8 data from the same reservoir was reviewed. Specifically, we aimed to identify the most suitable satellite-based spectral index for detecting chlorophyll-*a* in the Dongzhang Reservoir and to discern seasonal trends in chlorophyll content. Integrating satellite imagery with traditional sampling techniques will enhance our ability to effectively detect and monitor small algal blooms.

Materials methods

Study area

The Dongzhang Reservoir is the most important freshwater reservoir in Fujian Province, China. The Dongzhang Reservoir was built in 1957 and is located at the foot of Shizhu Mountain in the middle reaches of the Longjiang River in Fuqing city, Fujian Province, China (119°29'20" E, 25°46'6" N) (Fig. 1). Catchment area of 200 km² above the dam site. The Dongzhang Reservoir, with a resident population of

about 19,200, covers the Qingyuan community in Dongzhang Town, 15 administrative villages, and 5 villages in Jingyang Town. The area is surrounded by extensive agricultural land and has also been developed for tourism at the outlet. The reservoir is influenced by domestic sewage, agricultural runoff, and tourism-related activities, with nonpoint pollution being the predominant factor affecting water quality (Jin, 2021; Zahir et al., 2025). In 1992, the Fujian Provincial Government designated it as a drinking water source, requiring compliance with Class II water quality standards (GB3838-2002).

Currently, total phosphorus and nitrogen are the most significant pollutants, and the reservoir often shows mesotrophic conditions, with algal blooms reappearing in recent years (Weng, 2020a, 2020b; Zahir et al., 2025). Additional contamination originates from upstream rivers due to large-scale cage aquaculture, livestock and poultry farming, and untreated domestic sewage (Zhou, 2017). Sediment, accumulated over 65 years since the reservoir's construction, also serves as a nutrient source,

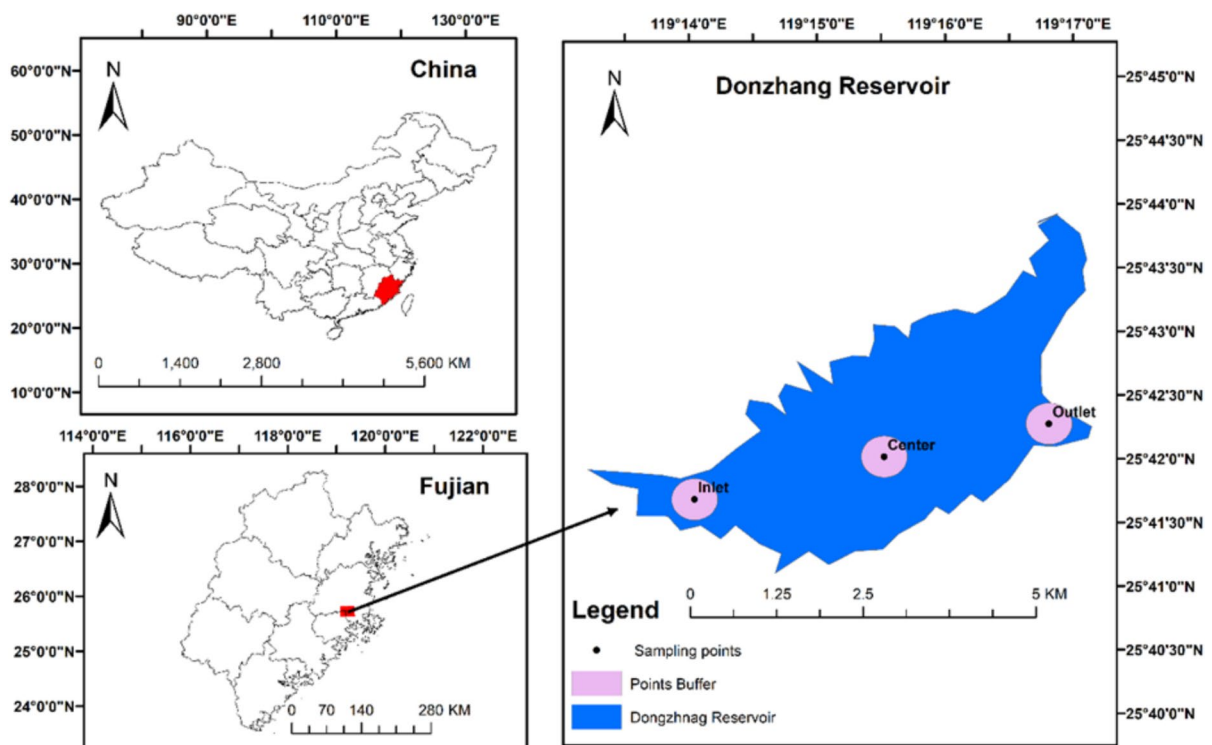


Figure. 1 Map showing Dongzhang Reservoir, Fujian, China, with in situ sampling points indicated by black dots surrounded by 300-meter buffers representing Sentinel-2 satellite data acquisition sites.

contributing to eutrophication (SU et al., 2008; Jin, 2021). Excessive nitrogen and phosphorus inputs primarily originate from agricultural, urban, and wastewater runoff, serving as the main drivers of eutrophication and subsequent algal proliferation (Li Jingjing et al., 2023). Variability in precipitation and inflow rates shows a strong correlation with nutrient concentrations. Altered rainfall regimes under climate change are likely to modify nutrient delivery patterns, thereby affecting both the onset and persistence of HABs (Lian et al., 2022).

Land-use expansion in the Dongzhang watershed has increased non-point source pollution, exposing the reservoir to greater agricultural runoff and domestic waste (Zahir et al., 2025). Sewage interception projects, untreated wastewater from villages, schools, hospitals, and military facilities continue to flow into the reservoir, as progress on pipeline connections has been slow. Villages such as Daqiao and Huashi are still discharged directly into the water. Livestock and poultry farms in nearby towns and villages further exacerbate nutrient loading, making animal husbandry a key driver of eutrophication (Liming, 2016; Environmental Protection Bureau of Fuqing). According to the catchment characteristics and geographic location of the Dongzhang Reservoir, the Fuzhou Monitoring Station established three long-term water quality sampling points. These stations are strategically positioned at the reservoir inlet (to capture upstream inflow and potential pollutant sources), the reservoir center (to represent overall water body conditions), and the outlet or dam site (to assess the quality of water being discharged downstream). This monitoring framework ensures a comprehensive evaluation of spatial variations in water quality within the reservoir system (Zahir et al., 2025). Variability in precipitation and inflow rates shows a strong correlation with nutrient concentrations. Altered rainfall regimes under climate change are likely to modify nutrient delivery patterns, thereby affecting both the onset and persistence of harmful algal blooms (HABs). The retention time, stratification, and inflow characteristics of Dongzhang Reservoir are influenced by water transfer projects, such as the Pingtan and Minjiang Estuary water resources allocation project. Longer retention times promote thermal stratification, leading to nutrient accumulation and increased algal growth, especially under nutrient-rich conditions (Li Jingjing et al., 2023).

In situ data collection

Three sampling stations were selected according to the reservoir area and the surrounding conditions of the reservoir: the inlet, center, and outlet (Fig. 1). The selection of these sampling points aims to capture critical hydrodynamic and anthropogenic influences on water quality: Inlet Station: Monitors upstream inflows and potential pollutant sources, such as agricultural runoff and domestic wastewater. Center Station: Represents overall water quality, capturing the distribution of nutrients and contaminants across the reservoir. Outlet Station: Assesses the water quality of downstream discharge, crucial for evaluating the impact of water transfer projects. Monthly samples were collected from March to December 2022. In May, June, July, August, September, and October, four samples were taken each month, corresponding to the active bloom period in Dongzhang Reservoir. In contrast, March, April, November, and December had only one sample each. The increased frequency during the bloom period provided more detailed insights into phytoplankton dynamics, as detailed in Table 2 (Zahir et al., 2025). A 1 L water sample from a depth of 0.5 m was fixed with 15 mL of Lugol's solution and preserved (Williams et al., 2016). After 48 h, the sample was concentrated to 30 mL using the siphon method. For counting, a 1 mL plankton chamber (2 mm×2 mm) was used, with 2 tablets counted per sample under a Leica microscope (Leica BM-E, Beijing) (Wu et al., 2019). Multiple fields of view were counted per chamber, and counts were averaged across replicates to reduce counting errors. Additionally, two years, 2022 and 2023, in situ water quality data were obtained from the Dongzhang automatic monitoring station and the Ministry of Ecological Environment, Fujian Provincial (municipal) Ecological Environment Bureau. (<https://sthjt.fujian.gov.cn/wsbs/bmfwcx/szcx/>). The data included algal density (10,000/L), chlorophyll-a concentration (µg/L), total phosphorus (TP) concentration (mg/L), total nitrogen (TN) concentration (mg/L), water surface temperature (°C), turbidity (NTU), dissolved oxygen (DO) concentration (mg/L), and pH. These parameters were chosen based on their established roles in HAB dynamics and their ability to provide a comprehensive understanding of water quality and algal growth

conditions. Missing values or inconsistencies in the dataset were handled using imputation or data interpolation techniques to ensure data integrity.

The identification of algae

The identification of algae was performed according to the Atlas of Common Algae in Chinese mainland Waters, which was calculated according to Eq. (1) (Zhang & Huang, 1991).

$$p = \frac{S}{S_0} \times \frac{V_0}{V} \times N_0, \quad (1)$$

In Eq. (1), N is the cell abundance (cells/L).

S is the area of the counting box (mm^2).

S_0 is the metric field area (mm^2).

V_0 represents the volume after concentration (mL).

V represents the count volume (mL).

N represents the number of algae per field count (cells).

To minimize uncertainties in microscopic counting, such as operator variability and counting errors, multiple counts per sample were averaged to ensure consistency and improve reproducibility. The statistical analysis and graphics were performed via Microsoft Excel (365) and OriginPro 9.0 software. Remote sensing data and in situ data were correlated via Pearson correlation analysis. Correlation analysis was performed via Pearson's p values (p values for significance between two variables in situ and index values). Prior to performing the correlation, data normality was assessed, and appropriate transformations were applied where necessary to meet the assumptions of normality.

Satellite data collection and processing and selection of spectral indices.

To monitor chlorophyll- a concentrations remotely, a variety of algorithms (spectral indices) have been developed (Johansen et al., 2018). Algal bloom information is mostly extracted from images via spectral indices since they are easy to implement and flexible (Cao et al., 2021). The infrared and visible bands are used to calculate spectral indices. Many studies have been conducted on developing spectral indices for monitoring algal blooms (Sagan et al., 2020; Shi et al., 2019; Xiong et al., 2020). Currently, several famous spectral indices are used for monitoring HABs, including the surface algal bloom index (SABI), floating algae index (FAI), algal bloom index (ABI), maximum chlorophyll index (MCI), normalized difference turbidity index (NDTI), surface scum index (SSI), normalized difference vegetation index (NDVI), normalized difference chlorophyll index (NDCI), scaled algae index (SAI), and red tide index (RI), among others. These indices have varying capabilities in distinguishing algal pixels from other pixels because of the different spectral bands and satellite data used in their algorithms. Therefore, choosing an appropriate index for differentiating algae in different environments with different satellite data is crucial (Alharbi, 2023).

To assess which spectral index best detects chlorophyll a at each sampling location, we applied the two spectral indices of Sentinel-2 for chlorophyll a identification (Table 1). The indices used in this study include the normalized difference vegetation index (NDVI) and normalized difference chlorophyll index (NDCI) (Table 1). Rrs indicates the reflectance values from the satellite. The wavelengths represented include the near-infrared (NIR), red, and vegetation red edge 1 (Table 1). To determine the spectral index values from the Sentinel-2 imagery, these images were clipped to a 300 m circular buffer surrounding each in situ sampling site (Fig. 1).

Table 1 Spectral indices used in this study for chlorophyll- a identification in the Dongzhang Reservoir

Algorithm	Equation	Wavelengths	Bands combination of Sentinel-2	References
NDVI	$Rrs842 - Rrs665 / Rrs842 + Rrs665$	NIR, Red	$(B8 - B4) / (B8 + B4)$	(Ma et al., 2021)
NDCI	$Rrs705 - Rrs665 / Rrs705 + Rrs665$	Veg red Edge 1, Red	$(B5 - b4) / (B5 + B4)$	(Caballero et al., 2020)

Sentinel-2 is a satellite mission developed by the European Space Agency (ESA) as part of the Copernicus program. Sentinel-2 sensors provide high-resolution multispectral imagery, making them ideal for monitoring HAB dynamics in reservoirs. The sensors capture images in the visible and near-infrared spectra, which allows for the detection of chlorophyll-a, a pigment found in algae (Liu et al., 2022; Mollae, 2018). Sentinel-2 provided us with multispectral imagery for our research locations during the study period. Sentinel-2A, which started running in 2015, and Sentinel-2B, which has been operational since 2017, are combined, offering high-resolution (10 to 20 m) images of the sample locations at 5-day intervals (Caballero et al., 2020). After processing and obtaining in situ data and phytoplankton densities in the Dongzhng Reservoir, we processed Sentinel-2 Level 1 images via the Google Earth Engine (GEE) (Fig. 2). It is a global-scale processing application programming interface for satellite imagery analysis (Lobo et al., 2021). The effectiveness of the Google Earth Engine for tracking inland water quality has been demonstrated in several recent studies (Jia et al., 2019; Lobo et al., 2021; Maeda et al., 2019; Vaičiūtė et al., 2021; Wang et al., 2020; Weber et al., 2020; Zong et al., 2019).

We selected Sentinel 2 images taken five days or less before the monthly in situ water quality collection dates from March to December 2022. Table 2 summarizes the details of the in situ sampling and satellite overflights and separates calibration and validation, which were produced to organize the data. Since the satellite overpass periods nearly coincided with the collection dates, field samples were used to establish site-specific connections between the NDVI and the NDCI and chlorophyll-a concentrations. We started monthly regular sampling in March, and the sampling frequency is given in Table 2.

Algal bloom proliferations in the Dongzhang Reservoir are highly variable due to their dynamic natures and nutrient inputs. All available Sentinel-2 images were collected to determine in situ sampling days (Table 2).

During the study period, most Sentinel-2 images were usable, as Dongzhang Reservoir generally experiences favorable weather during spring and summer. Cloud contamination only affected a limited number of images, which were excluded using the QA60 band mask. The dataset underwent masking with a QA60 (quality assurance) band to eliminate dense and cirrus clouds. Since analysis relied on monthly sampling windows, at least one valid image per period.

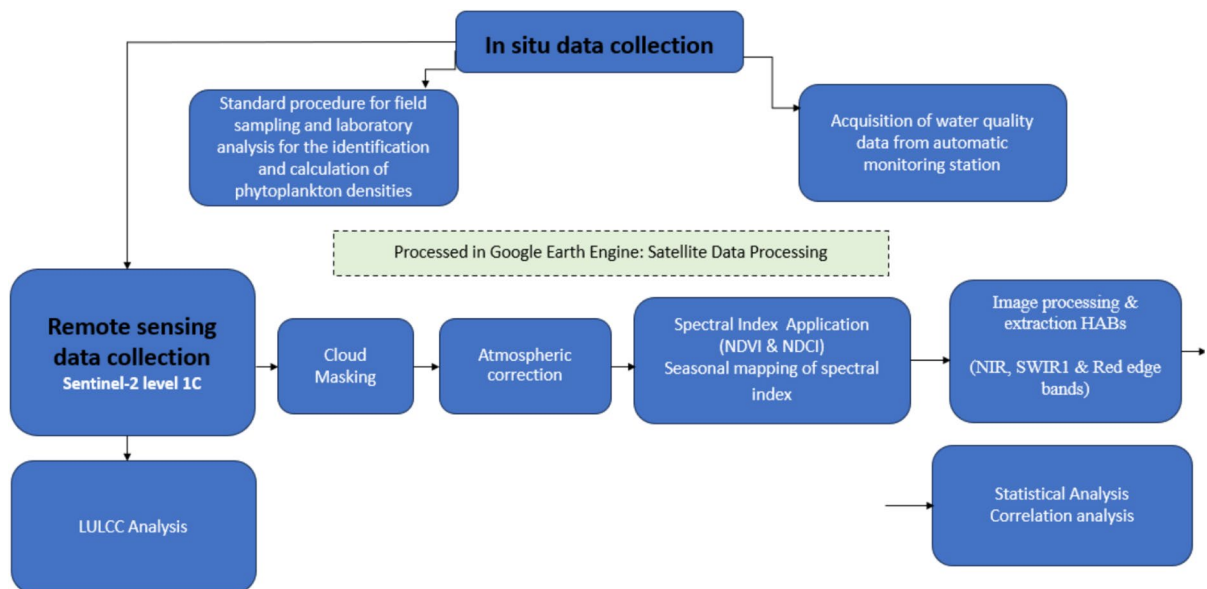


Fig. 2 Diagram illustrates the process of acquiring, processing, and analyzing satellite images and in situ data

Table 2 Details of the remote sensing data, in situ data values and sampling dates of the Donzhang Reservoir in 2022

RS (index values)			Days since sample		In situ (total phytoplankton density C/L)			
Sentinel date	NDVI	NDCI			Sample date	Inlet	Centre	Outlet
4-Mar-22	0.055	0.081	1	3-Mar-22		2.11E+05	6.46E+06	6.46E+06
14-Mar-22	0.043	0.074	1	15- Mar-22		0.00E+00	8.56E+08	9.36E+08
23-Apr-22	0.02	0.021	4	27-Apr-22		0.00E+00		3.55E+08
8-May-22	0.039	0.103	1	7-May- 22		1.36E+08	1.27E+08	1.04E+08
13-May-22	0.021	0.083	0	13-May-22		8.60E+07	7.40E+07	5.80E+07
18-May-22	0.09	0.113	2	20-May-22		9.00E+07	8.05E+07	6.11E+07
28-May-22	0.001	0.285	1	27-May-22		5.85E+07	4.24E+07	3.34E+07
2-Jun-22	0.067	0.035	0	2-Jun-22		6.87E+07	5.74E+07	3.67E+07
12-Jun-22	0.021	0.017	3	9-Jun-22		4.69E+07	2.73E+07	4.10E+07
17-Jun-22	0.018	0.238	1	16-Ju-22		4.10E+07	6.86E+07	5.05E+07
22-Jun-22	0.056	0.078	1	23-Jun-22		5.61E+07	7.33E+07	6.34E+07
27-Jun-22	0.068	0.082	3	30-Jun-22		6.34E+07	7.17E+07	4.42E+07
7-Jul-22	0.041	0.051	0	7-July 22		4.55E+07	4.29E+07	5.16E+07
12-Jul-22	0.1	0.093	2	14-July 22		3.71E+07	3.09E+07	2.86E+07
22-Jul-22	0.064	0.061	1	21-July 22		2.86E+07	4.30E+07	3.77E+07
27-Jul-22	0.046	0.061	1	28-July 22		2.76E+07	3.97E+07	3.08E+07
6-Aug-22	0.073	0.076	1	5-Aug-22		3.03E+07	2.54E+07	1.71E+07
11-Aug-22	0.08	0.075	0	11-Aug-22		3.85E+07	2.49E+07	3.46E+07
16-Aug-22	0.083	0.098	2	18-Aug-22		3.56E+07	3.29E+07	4.28E+07
26-Aug-22	0.041	0.177	0	26-Aug-22		4.28E+07	3.61E+07	3.56E+07
5-Sep-22	0.048	0.139	4	1-Sep-22		3.35E+07	4.15E+07	3.24E+07
10-Sep-22	0.022	0.028	2	8-Sep-22		5.38E+07	6.91E+07	6.06E+07
15-Sep-22	0.049	0.092	0	15-Sep-22		4.20E+07	5.89E+07	4.39E+07
20-Sep-22	0.03	0.033	1	21-Sep-22		3.30E+07	4.15E+07	3.04E+07
30-Sep-22	0.052	0.069	1	29-Sep-22		3.91E+07	1.99E+07	2.49E+07
10-Oct-22	0.039	0.107	4	6-Oct-22		3.34E+07	2.32E+07	1.22E+07
15-Oct-22	0.021	0.074	2	13-Oct-22		2.30E+07	7.65E+06	1.67E+07
20-Oct-22	0.023	0.066	0	20-Oct-22		3.02E+07	2.09E+07	2.74E+07
25-Oct-22	0.016	0.041	2	27-Oct-22		2.42E+07	3.28E+07	1.76E+07
4-Nov-22	0.021	0.006	1	3-Nov-22		2.68E+07	2.38E+07	1.15E+07
4-Dec-22	0.046	0.064	1	5-Dec-22			1.41E+07	1.26E+07

We collected $31 \times 3 = 93$ samples from three locations within the Dongzhang Reservoir between March and December 2022. Nearly two-thirds (20 out of 31) of these images were taken between 0 and 1 day after in situ collection. Six out of 31 samples occurred on the same day as the in-situ sample date

Therefore, missing data did not significantly affect the analysis.

The dataset underwent masking with a QA60 (quality assurance) band to eliminate dense and cirrus clouds. Employing the satellite invariant atmospheric correction (SIAC) package, a satellite-based atmospheric correction algorithm was used to convert top-of-atmosphere images into surface reflectance images. This package integrates Bayesian statistics and the Copernicus Atmospheric Monitoring

Service (Yin et al., 2019), employing linear transformations to estimate surface reflectance and aerosol optical thickness. In their study, Lobo et al. (2021) applied this atmospheric correction algorithm to Sentinel-2 images within the GEE to derive NDCI values. Their results indicated a strong correlation ($R^2 = 0.96$) between this algorithm and Sentinel-2 surface reflectance data, particularly post-2019 data (Lobo et al., 2021).

In situ phytoplankton density measurements were correlated with spectral index values on the basis of previous studies (Konik et al., 2020; Kwon et al., 2018; Martinez et al., 2020; Sharma et al., 2019). A median pixel is created for the NDVI and NDCI by comparing the seasonal spectral index values against the total phytoplankton density in the Dongzhang Reservoir. The median values are less sensitive to extreme reflectance values caused by floating debris, shadows, or sensor noise than mean. This helped to reduce outlier effects. Cloud masking using the QA60 band, and the use of median pixel values helped mitigate localized noise from suspended particles. To better understand the temporal patterns of blooms, this seasonal analysis is helpful. Additionally, we used Landsat 8 data to create a water surface temperature map and compare in situ temperatures with remote sensing values at the Dongzhang Reservoir.

Results and discussions

Temporal variation in spectral indices

We conducted an analysis of the spatial and temporal variations in the NDVI and NDCI (median pixel values) within the Dongzhang Reservoir during the year 2022. Considering the seasonal variations in spectral index values observed in our study, we defined the seasons as follows: March to May, representing spring in Fujian; June to August, denoting summer; September to November, representing autumn; and December to the following February, representing winter. We selected these three seasons (Fig. 3) because they coincide with the periods during which harmful algal blooms (HABs) are typically observed in the Dongzhang Reservoir (Chen, 2020). Using Sentinel-2 satellite imagery and spectral indices, we identified patterns indicative of algal blooms within the Dongzhang Reservoir (Fig. 3). Our findings revealed that algae levels

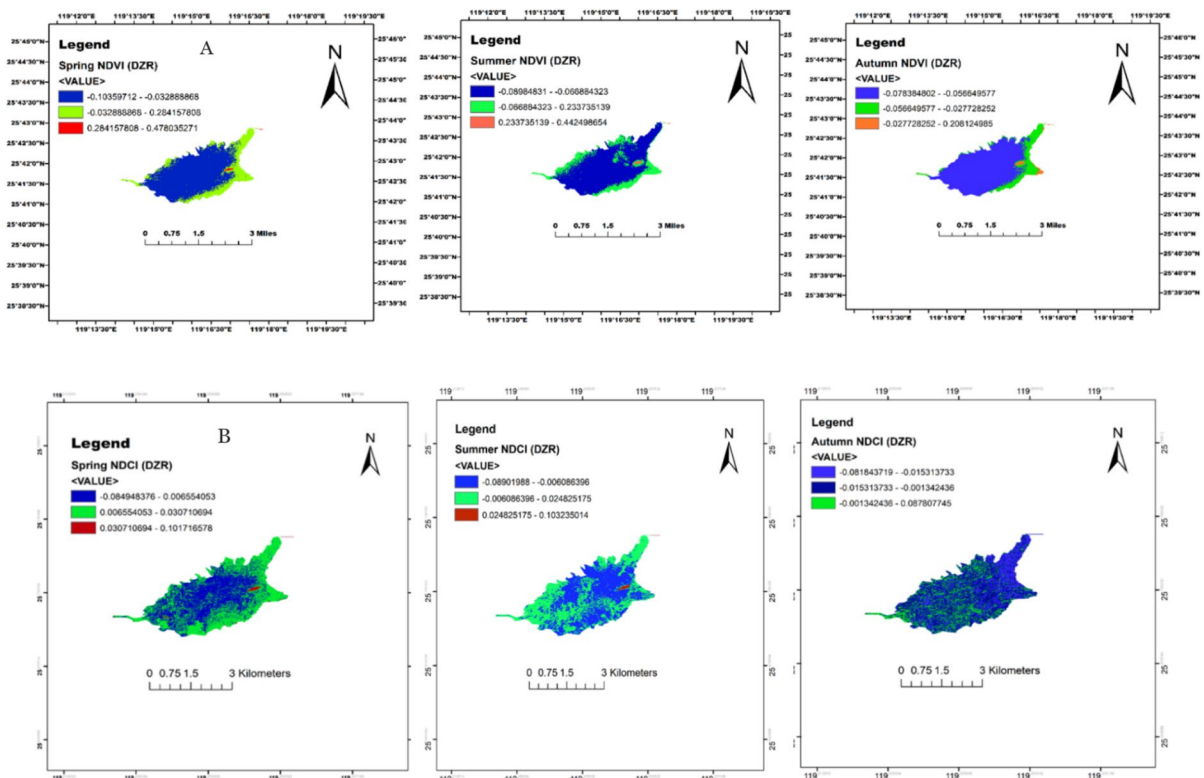


Figure 3 Analysis of the temporal variation in the NDVI (A) and NDCI (B) (seasonal median pixel values) in the Dongzhang Reservoir in 2022

were highest during the spring and summer months, with the peak occurring in May. Additionally, elevated levels of algal blooms were observed in the Dongzhang Reservoir in March 2022.

As early as autumn, some chlorophyll had already appeared. It was more prevalent in Dongzhang Reservoir. Sentinel-2 imagery demonstrated that the NDCI consistently estimates chlorophyll in the Dongzhang Reservoir (Fig. 3B). The NDCI also revealed that chlorophyll concentrations are spread over a wide area of the reservoir (Fig. 3B). A positive relationship was observed between both the NDCI and the NDVI, with the NDVI demonstrating a relatively high chlorophyll concentration at the edge of the reservoir (Fig. 3A). In spring and summer, the NDCI values were greater. NDCI has

shown up to ~20% higher correlation than NDVI quantifying the relative improvement in R^2 values.

Linear regression analysis between phytoplankton density and remote sensing values in spring, summer and autumn 2022 for the Dongzhang Reservoir is shown in Fig. 4. The relationships between the in-situ measurements and index values were significant (Fig. 4abc). A positive correlation occurs during spring ($R^2=0.81$), and the p value is highly significant (<0.0001) (Fig. 4a), whereas during summer ($R^2=0.74$), the p value <0.0002 (Fig. 4b), and in autumn ($R^2=0.62$), the p value is <0.0001 (Fig. 4c). Both indices performed well at the Dongzhang Reservoir, but the NDCI performed best. A comparison of both indices is shown in Fig. 5. In May, the value was greater (Fig. 5). By combining sensors for NIR

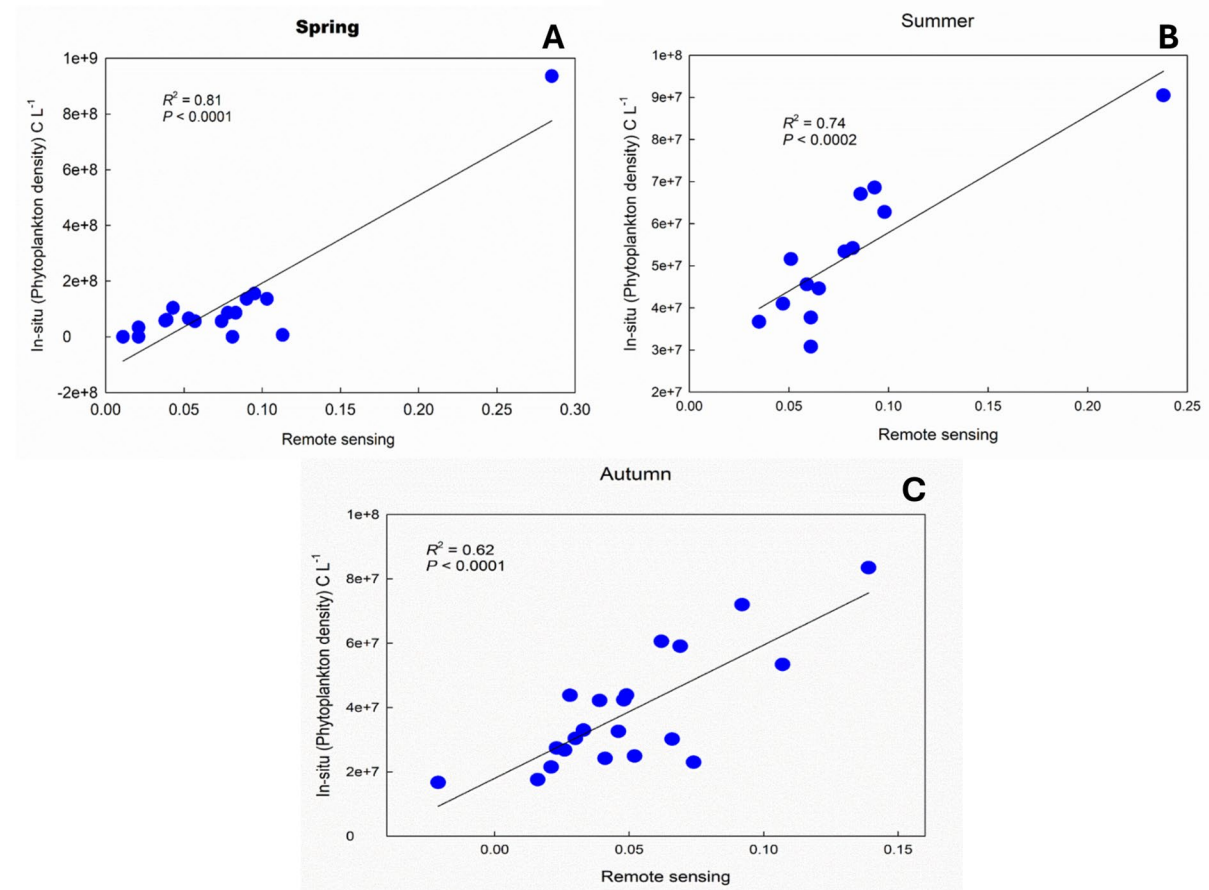


Figure. 4 Linear regressions between in situ data and remote sensing values in spring, summer and autumn 2022 of the Dongzhang Reservoir

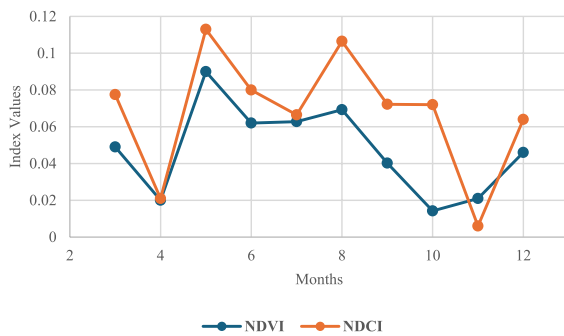


Figure.5 Comparison of both spectral indices (NDCI & NDVI) from March to November 2022 in the Dongzhang Reservoir

and red (R) wavelengths, (Rouse, 1973) developed the NDVI for detecting vegetation cover, and the equation can be seen in Table 1. Microalgae have a threshold value of the NDVI of -0.15 , whereas water has values close to zero (-0.1×0). A high level of chlorophyll-a can lead to increased NDCI, which can be used to monitor algae blooms (Lobo et al., 2021).

The NDCI, which is a standardized algal index, was proposed by Mishra and Mishra (2012) on the basis of remote sensing data from turbid productive waters. This method exploits the reflectance peak of chlorophyll-a at 708 nm and the strong absorption characteristic of red chlorophyll-a (665 nm) to calculate chlorophyll-a concentrations.

$$NDCI \propto [Rrs(708) - Rrs(665)] / [Rrs(708) + Rrs(665)]$$

Using the ESA's medium resolution imaging spectrometer (MERIS) sensor, Mishra and Mishra (2012) demonstrated a robust correlation between the NDCI and chlorophyll-a concentration, yielding an R^2 of 0.95 with a mean standard error of 2.49 mg m⁻³ ($p < 0.0001$). To build upon this correlation, chlorophyll concentrations were derived via a quadratic function applied to the generated NDCI Multi-Spectral Instrument (MSI) images. To mitigate variations arising from solar and atmospheric factors, the NDCI leverages the spectral band difference between 709 and 685 nm, normalized by the sum of the two reflectance values (Mishra & Mishra, 2012). These specific spectral bands were selected for algal bloom detection because of their historical efficacy in identifying chlorophyll-a.

Indices based on red, red-edge, and near-infrared wavelengths excel in turbid waters, as they are less influenced by surrounding debris and organic matter, and those utilizing blue and green wavelengths perform optimally in clear water bodies. Given the diverse natures of the studied reservoirs, both types of indices were explored in this investigation.

According to previous studies, the two-band model was superior to the three-band model at estimating chlorophyll a. In an inland tropical reservoir, Augusto-Silva et al. (2014) evaluated the performance of two- and three-band NIR-red models with NDCIs at close ranges and discovered that the NDCI was the most accurate model. By simulating reflectance in various spectral bands of several sensors, Beck et al. (2017) compared 12 bio-optical models used for the assessment of chlorophyll-a concentrations in Harsha Lake, OH, USA. NDCI is the model with the broadest applicability and the best performance for most inland water remote sensing sensors, including MERIS, WorldView-2, MSI Sentinel-2, and OLCI Sentinel-3 (Ogashawara et al., 2017).

The results for the Dongzhang Reservoir show that Sentinel-2 imagery is a good tool for monitoring small inland water bodies. Strong correlations between in situ data and satellite-derived spectral indices were found at each site in the Dongzhang Reservoir, where we collected water samples from three separate locations, particularly in the inlet region (Table 2), which provides more information. While the matching of Sentinel-2 images to in situ sampling dates is well-explained in (Table 2), temporal mismatches in months with rapid bloom dynamics could influence the correlation between spectral indices and chlorophyll-a concentrations. For instance, in June and July 2022, rapid changes in algal density (e.g., $6.86E+07$ cells/L on 16-Jun-22) may not align with satellite data from 23-Jun-22, potentially affecting the accuracy of the correlation. Temporal proximity of sample and satellite data was considered to minimize these effects.

Temporal dynamics of HABs in the dongzhang reservoir

Seasonal maps illustrate the temporal dynamics of HABs in the Dongzhang Reservoir (Fig. 3). The range of "water blooms" spread to the reservoir area during spring, summer, and autumn, as shown in

Fig. 3. According to the seasonal analysis, spring and summer are the two seasons with the highest concentrations of chlorophyll-a, particularly near the Qixingchi village side, Shangli village side and Maba village side, as well as the inlet and all tourists' offshore points. A high concentration of chlorophyll-a was also detected where river tributaries were located. Several villages surround the Dongzhang Reservoir, an even greater amount of agriculture is being grown there, and the outlet of the reservoir is a tourist development area (Zahir et al., 2025; Chen, 2020). Living and agricultural pollution around Dongzhang Reservoir, as well as some tourism development activities, pollutes the reservoir (Wang, 2022; Chen, 2020).

In a temporal standardized index map, the relative difference between each spectral index value is shown across space and time. According to these maps, diverse locations inside the reservoir have different tendencies in outbreaks of HABs depending on the time. The NDVI in Dongzhang Reservoir differed across locations; for example, in autumn, the NDVI did not reveal HABs at the inlet, but at the outlet, the concentration was high (Fig. 3). Thus, these maps can be used to determine the seasonal and localized spatial patterns of algal blooms on the basis of satellite data. The HABs in the Dongzhang Reservoir show obvious seasonal variations, and vertical stratification is obvious in spring and summer. That is, the surface layer of algae erupts in spring and summer. In accordance with the distribution law of algae, the discharge of algae before the flood season should be considered in reservoir operation. Moreover, the elevation of water intake can be adjusted in time by using a changing law to reduce the effect of algae blooms on the water supply quality of water (Chen, 2020). Cyanobacteria, green algae, and diatoms make up most of the phytoplankton in the Dongzhang Reservoir, with cyanobacteria being the largest category. Five phyla and 19 genera of phytoplankton were found, with the density of phytoplankton ranging from 2.06×10^7 to $5.45 \times 10^7 \text{ L}^{-1}$ in the summer of 2021 (Wang et al., 2022). The Dongzhang Reservoir has a large amount of pollution, and multiple environmental conditions have an impact on phytoplankton density (Chen, 2020). These findings demonstrate that Dongzhang Reservoir's state of health was worse in the summer of 2021 than it was in the summer of 2020 (Wang et al., 2022).

The main "bloom" species are *Microcystis*, *Anabaena spiralis* and filamentous algae in the Cyanophyta phylum. *Chlorella vulgaris* also accounts for a considerable proportion. Occasional outbreaks of HABs in Dongzhang Reservoir have been reported (Dying, 1995). In March and November, the amount of rainfall was low, the temperature changed greatly, and the temperature fluctuated greatly, which affected the propagation of algae (Chen et al., 2023; Zhen, 2017).

A consistent variation trend of phytoplankton biomass was found in the inlet, center, and outlet sections of the reservoir, with the peak occurring in May. The inlet section had the highest phytoplankton biomass of the three sections studied (Fig. 6). The phytoplankton in the Dongzhang Reservoir are composed mainly of cyanobacteria, green algae, diatoms, cryptoalgae, and a few other algae. Figures 2, 3, 4, 5, 6, 7, 8, 9 show the dominant phytoplankton species in the month of May 2022. Cyanobacteria was the most dominant species in the inlet, center, and outlet $8.60\text{E}+07$, $7.82\text{E}+07$, and $5.75\text{E}+07$ cells/L, respectively. The MC density was also greater at points $7.79\text{E}+07$, $7.02\text{E}+07$ and $4.78\text{E}+07$ (Fig. 7). *Atuynia* was also higher in the inlet section, with $8.11\text{E}+06$ cell/L. The results of this study suggest that to prevent eutrophication of the Dongzhang Reservoir, attention should be given to cyanobacteria outbreaks during spring and summer, monitoring phytoplankton growth in the inlet section, and severely limiting nutrient inputs. There are interannual variations in the chlorophyll mass concentration, which are greater in summer than in spring and winter (Zhen, 2017). Cyanophyta, Chlorophyta, Bacillariophyta, Cryptophyta and Pyrophyta were detected in the wet season of the Dongzhang Reservoir in 2020, and the community structure was dominated by Cyanophyta. The average total phytoplankton density ($2.3 \times 10^7 \sim 5.2 \times 10^7$ cells/L) at the outlet of the dam during the wet season in 2021 was the same as that during the dry season ($5.6 \times 10^6 \sim 2.3 \times 10^7$ cells/L) (Chen et al., 2023). The mass concentration near the coast and inlet was higher than that in the reservoir center, which roughly reflected the temporal and spatial changes in the chlorophyll-a mass concentration (Zhen, 2017). In addition, the Dongzhang Reservoir should complete the sewage interception project of domestic sewage and farmland fertilization irrigation along the coast as soon as possible, which can

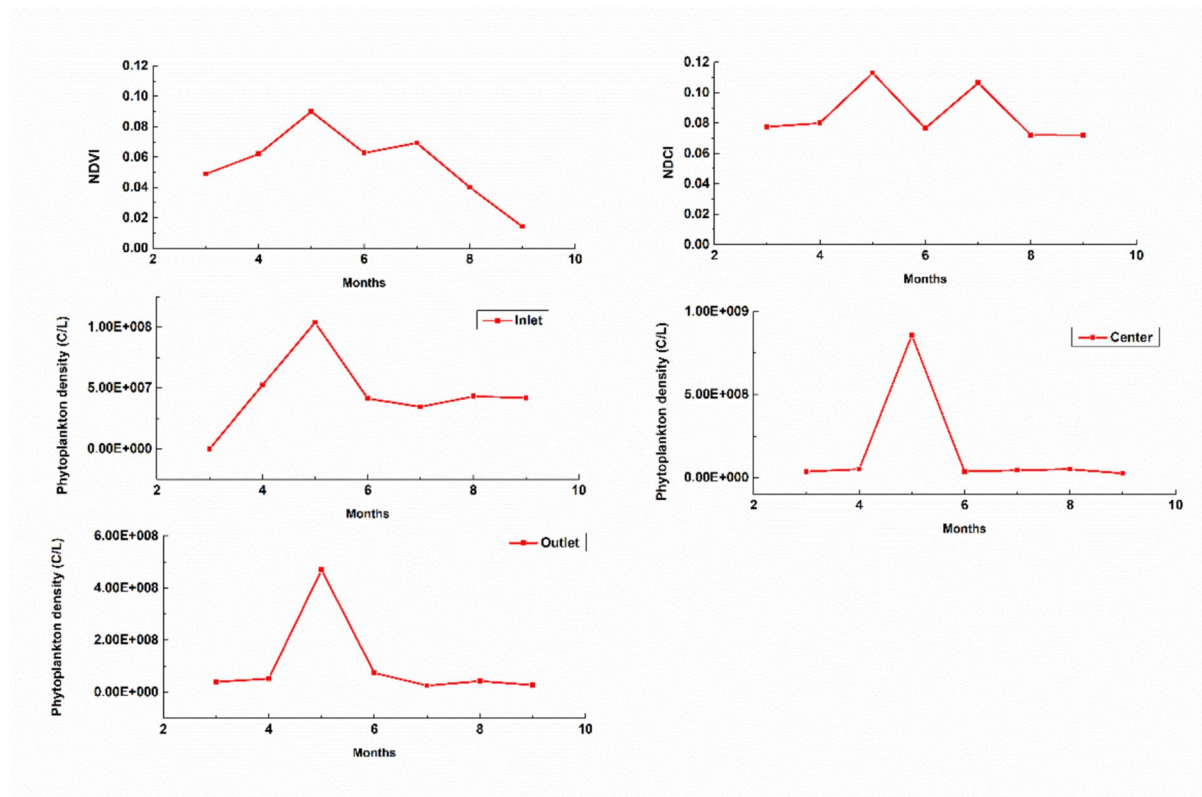


Figure. 6 Temporal variation trends of the NDVI, NDCI and phytoplankton biomass in the inlet, center, and outlet sections of the Dongzhang Reservoir

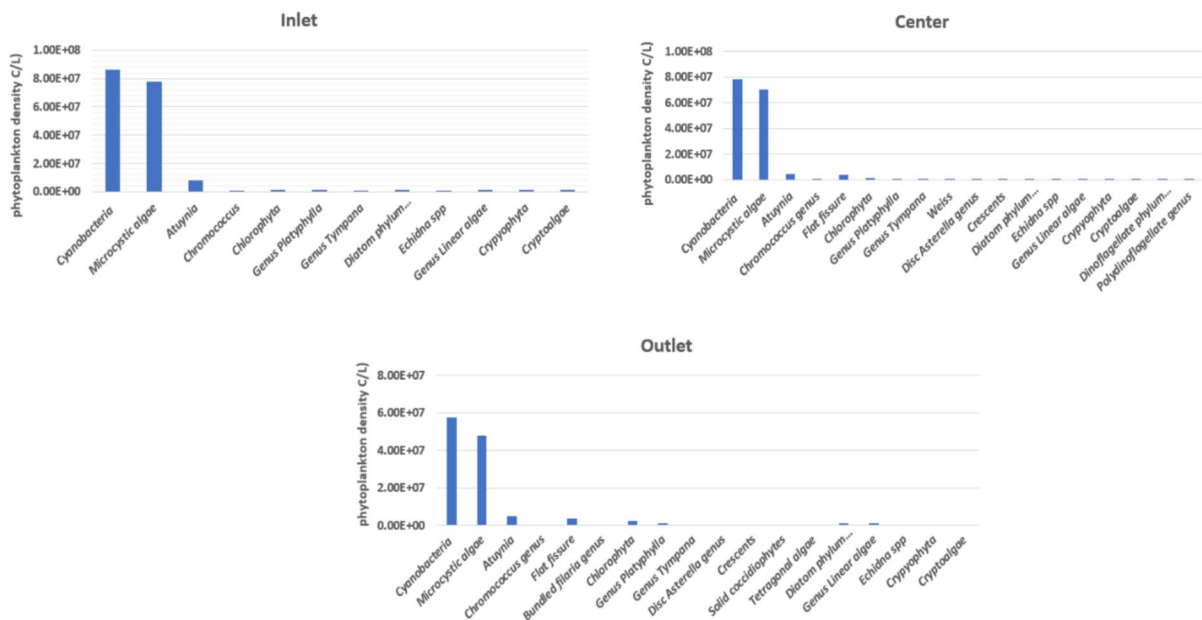


Figure. 7 Dominant phytoplankton species and their densities at three sampling locations in the Dongzhang Reservoir in May 2022

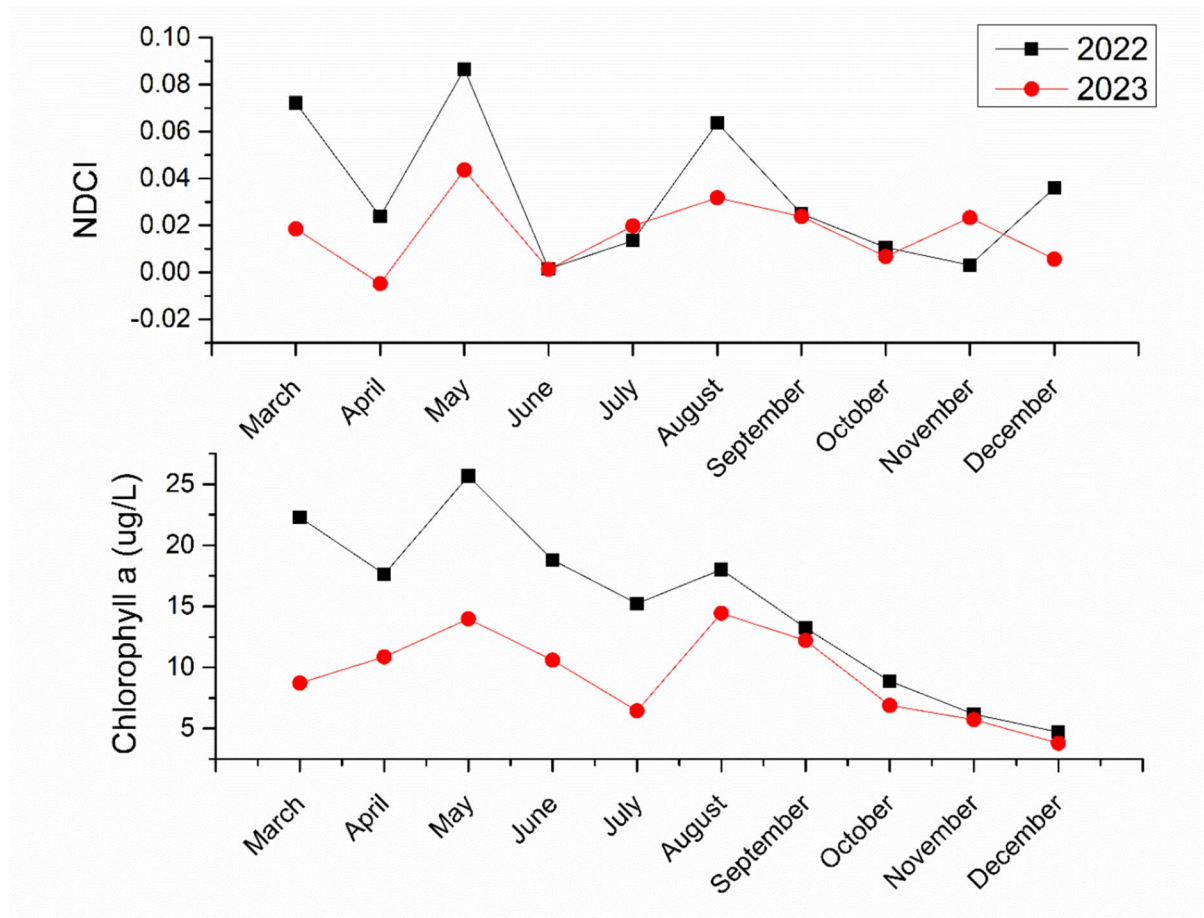


Figure. 8 Comparison of in situ chlorophyll values and remotely sensed chlorophyll values (NDCIs)

greatly reduce the current serious eutrophication risk of the Dongzhang Reservoir in the future; additionally, attention should be given to monitoring upstream water quality during high outbreaks of algae in summer and autumn and, in the future, water withdrawal, water transfer and more in-depth research on the Dongzhang Reservoir. The impacts on HABs in terms of water quantity and hydrodynamics should be supplemented, which is highly important for ensuring a safe water supply for the water source of the Dongzhang Reservoir in the future. The goal of eutrophication prevention and control can only be realized after a long period of comprehensive treatment with scientific ecological restoration measures based on effective control of nutrient pollution loads (Zhuang et al., 2013). The Dongzhang Reservoir is on the southeastern coast, the temperature and precipitation conditions are suitable for plant survival, and the

interannual distributions of chlorophyll mass concentrations differ (Zhen, 2017). The reason is that the rainfall in summer is abundant, and the temperature of the water body is stable, which is suitable for algae growth. From the end of spring, a small amount of bloom appeared on the reservoir surface of the dam site section in the downwind area (Chen, 2020; Wang et al., 2022).

With increasing water temperature, the number of "blooms" peaked in summer and autumn, and the scope of "blooms" spread widely to the reservoir area. The main types of blooms are *Microcystis* alga, *Anabaena spirulina* and *Syringoides* alga. *Oscillatoria* and *Chlorella* also accounted for a considerable proportion. In recent years, local eutrophication has occurred occasionally in the Dongzhang Reservoir (Chen, 2020).

In November, less rainfall occurs, the temperature changes dramatically, and temperature fluctuations affect algae reproduction (Zhen, 2017). Figures 2, 3, 4, 5, 6, 7, 8 also show that the values of the NDVI and NDCI were lower in November at the study site.

Dynamic comparison of HABs between 2022 and 2023

We compared the dynamics of HABs between 2022 and 2023. Overall, Dongzhang Reservoir was affected by severe HAB events in 2022. Figure 8 shows a comparison of 2022 and 2023 in situ chlorophyll data with spectral index values. The chlorophyll-a concentration was higher in 2022 and lower in 2023 than in 2022. A monthly comparison of the index values (Fig. 9) revealed that the average NDVI value was 0.049 in March 2022, whereas it was 0.040 in March 2023. We also compare the NDCI values in May 2022 and May 2023 (Fig. 10). The NDCI value in May 2022 was also higher than that in 2023, and we also calculated the area affected by HABs in the reservoir in May. In May 2022, HABs expanded across the reservoir, and a 4.8942 km² area was affected by the HAB event, accounting for 34.41% of the area of the reservoir; however, in May 2023, a 3.15 km² area was affected by the HAB, accounting for 22.15% of the reservoir area (Fig. 10).

Threshold

Spectral indices are typically used to distinguish between algal and water cells by determining a threshold value. Thresholds are determined empirically by referencing values from previous HAB studies and through visual inspection of spectral index histograms across seasons. This approach provided consistency with earlier work while ensuring adaptability to local reservoir conditions. The threshold values were adjusted to reflect the specific turbidity, CDOM, and algal coverage conditions of Dongzhang Reservoir, rather than adopting a single universal threshold. These threshold values vary depending on the water conditions, such as clear water, turbid water, and turbid water mixed with algae, which means that they differ from one body of water to another. Using spectral indices, the threshold value was determined to be -0.05 in the Red Sea (Alharbi, 2023). Pu et al. (2022) conducted a study on Chinese lakes via

Sentinel 2 images and established threshold values for the NDVI, FAI, and NDWI. They classified a point as an algal bloom if the $NDVI > -0.1$, $FAI > 0.003$ or $NDWIR-SWIR > 0$, with thresholds for different lakes ranging from 0.5–0.85. Another study focused on Laguna Lake in the Philippines (Caballero & Navarro, 2021) and reported that the daily percentage of pixels covered by blooms ($NDCI \geq 0.2$) indicated its presence. Figure 3 in this study shows seasonal maps displaying the threshold values for the NDVI and NDCI. In Dongzhang Reservoir, the NDVI thresholds varied from -0.032 in spring to -0.066 in summer and -0.056 in autumn, whereas the NDCIs in spring, summer, and autumn in Dongzhang Reservoir were 0.006, 0.006 and -0.015, respectively (Fig. 3).

Factors affecting the proliferation of HABs in the Dongzhang Reservoir

There are many factors that affect the growth of algae in water reservoirs, such as natural factors such as geographical location and climatic conditions, as well as environmental factors such as light, temperature, pH, nitrogen, phosphorus, etc. (Ayub et al., 2024; Akhtar et al., 2021; Bhateria and Jain., 2016; Li et al., 2023). A linear regression analysis between the algal density and several water quality parameters, i.e., chlorophyll-a ($\mu\text{g/L}$), total phosphorus (mg/L), total nitrogen (mg/L), dissolved oxygen (mg/L), water surface temperature ($^{\circ}\text{C}$), turbidity (NTU), and pH, was conducted for 2022 and 2023 (Fig. 11) and (Fig. 12). These water quality parameters and algal density were positively correlated. Total phosphorus and water surface temperature were the main factors affecting (R^2 0.83) and (R^2 0.81), respectively, phytoplankton density in 2022 at Dongzhang Reservoir (Fig. 11). In 2023, correlation analysis revealed that TP was the main factor affecting phytoplankton density in the Dongzhang Reservoir (Fig. 12).

The eutrophication that occurs in surface waters is often associated with nutrient factors, especially nitrogen and phosphorus. (Devlin & Brodie, 2023; Soro et al., 2023). Studies have shown that approximately 80% of lake eutrophication is caused by phosphorus constraints, approximately 10% is related to nitrogen, and approximately 10% is related to other factors (Smith et al., 2016). Different conditions, such as various natural geographic conditions, seasons, light intensities, nutrient compositions, and

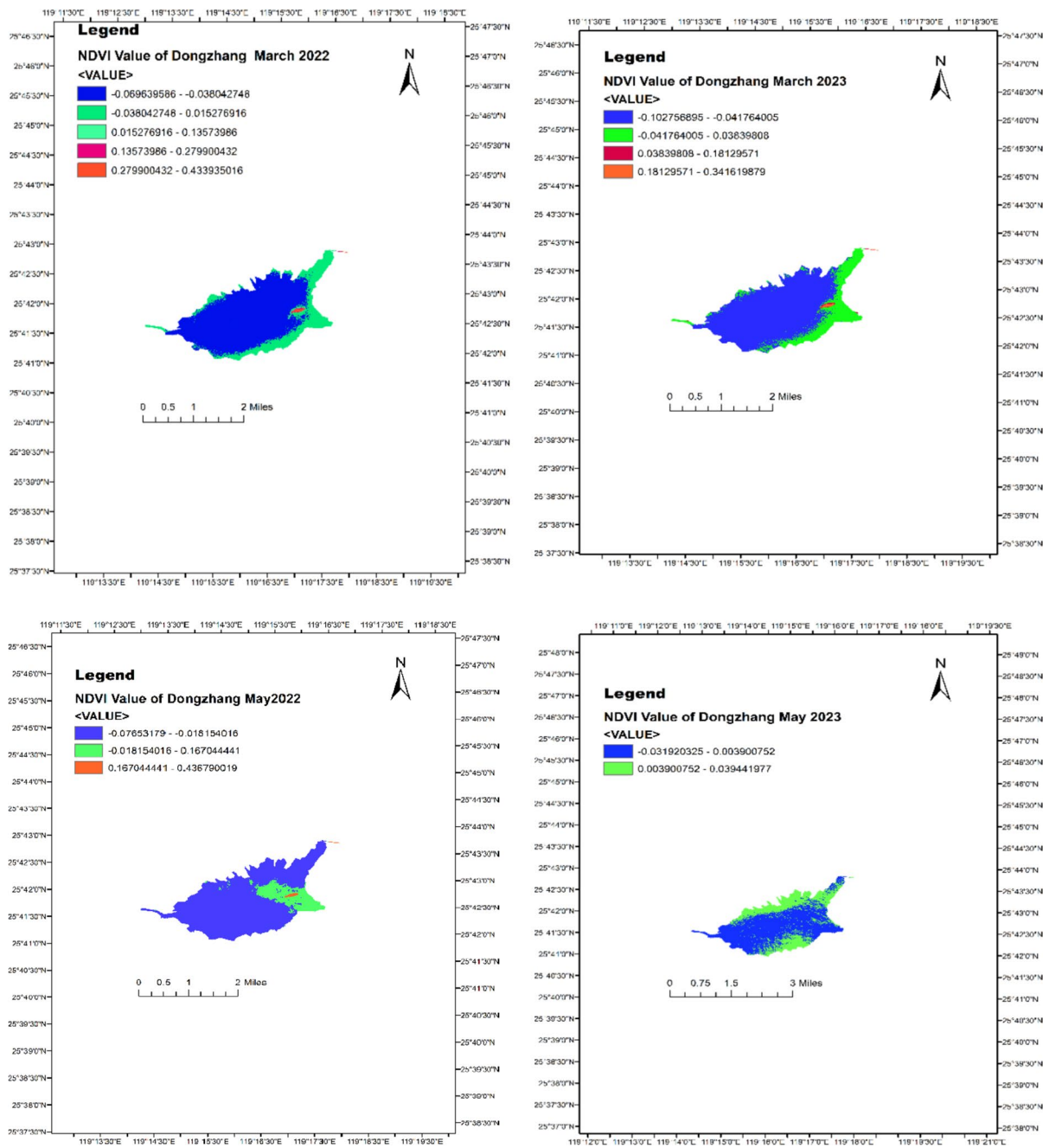


Figure. 9 Comparison of the NDVI values between March and May 2022 and 2023 for the Dongzhang Reservoir

eutrophication conditions, present very different characteristics. Our study also highlighted the importance of the Dongzhang Reservoir, which is the surface water head site in Fuqing city. In recent years, algal blooms have also occurred in different seasons. Research on the key factors affecting algae growth

in the Dongzhang Reservoir can provide a reference for the operation and management of the reservoir and is highly important for ensuring the safety of urban drinking water. Agricultural and living activities surround the reservoir, which creates a risk of water pollution (Liu et al., 2021). Open waterways

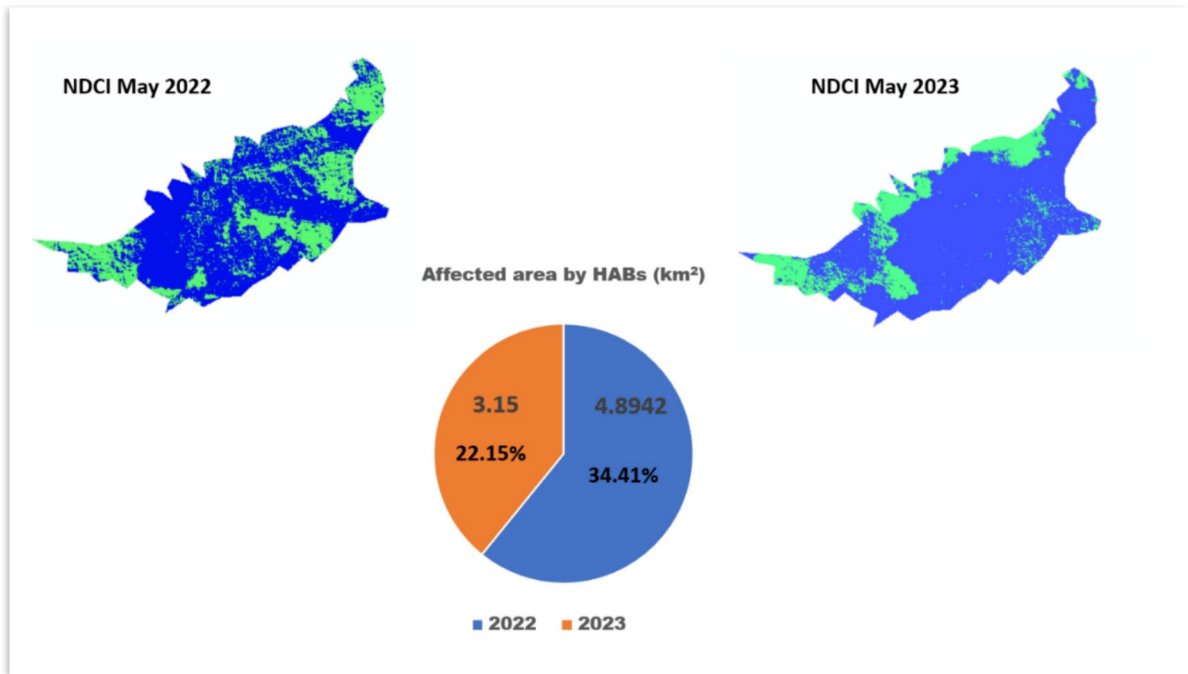


Figure.10 NDCI comparison between May 2022 and May 2023 (the green color represents HABs) and a pie chart showing areas affected by HAB events in May 2022 and May 2023

are constantly surrounded by farms or residential areas, increasing the potential for water pollution. To achieve this goal, we created a land use/land change cover map of the Dongzhang reservoir catchment. The LULCC map (Fig. 13) shows cropland around the reservoir. Agricultural activity is the most widespread source of impact in subtropical reservoirs (Burford et al., 2007). Many agricultural activities have been reported around the Dongzhang Reservoir (Chen, 2020). Many factors contribute to the degradation of water quality, including vegetation cover, climate change, river topography, and land use (Lintern et al., 2018). Nutrient enrichment, particularly from nitrogen and phosphorus, is a primary catalyst for HAB occurrence. Although essential for plant growth and natural aquatic ecosystems, human activities substantially increase their levels in water bodies. Sources include municipal and industrial wastewater discharges, concentrated animal feeding operations (CAFOs), urban stormwater runoff, septic system malfunctions, nitrogen deposition from fossil fuel emissions, and runoff from fertilized croplands and animal feeding operations. In Dongzhang Reservoir, total phosphorus and total nitrogen were also found

to be critical factors affecting phytoplankton density. Another study revealed the same results: total phosphorus and total nitrogen influenced phytoplankton density in the Dongzhang Reservoir (Liming, 2016).

Generally, chlorophyll-a can be a good indicator of algal blooms in water reservoirs (Yang et al., 2020). Chlorophyll-a is a valuable index for evaluating phytoplankton biomass (Li et al., 2022; Zahir et al., 2023). High chlorophyll concentrations suggest increased phytoplankton density, which can impact water quality and ecosystem health. Our results revealed that there was a positive correlation between algal density and chlorophyll concentration (Fig. 11c). Figure 8 shows the monthly variation in chlorophyll concentration, which was greater in the Dongzhang Reservoir than in the other reservoirs during May and August. The overall chlorophyll concentration was relatively high in 2022 and relatively low in 2023. The turbidity value is a measure of how much suspended matter blocks light from passing through water. High turbidity can result from various factors, including sediment, organic matter, and algal cells. Algal blooms, on the other hand, occur when the population of algae in water bodies rapidly

increases. In water reservoirs, elevated turbidity can be associated with algal blooms. Algal blooms contribute to increased organic matter and particles in the water, affecting its clarity. Monitoring both turbidity and algal bloom occurrence is crucial for assessing water quality, as excessive algal growth can lead to issues such as reduced oxygen levels, taste and odor problems, and potential harm to aquatic ecosystems. The relationship between algal density and turbidity (Fig. 11G) shows that algal blooms in reservoirs can significantly impact dissolved oxygen levels. During an algal bloom, the rapid growth of algae can lead to increased oxygen consumption, especially at night, when photosynthesis decreases. Additionally, the decay of excessive algal biomass can deplete dissolved oxygen as microorganisms breakdown organic matter (Brooks et al., 2016).

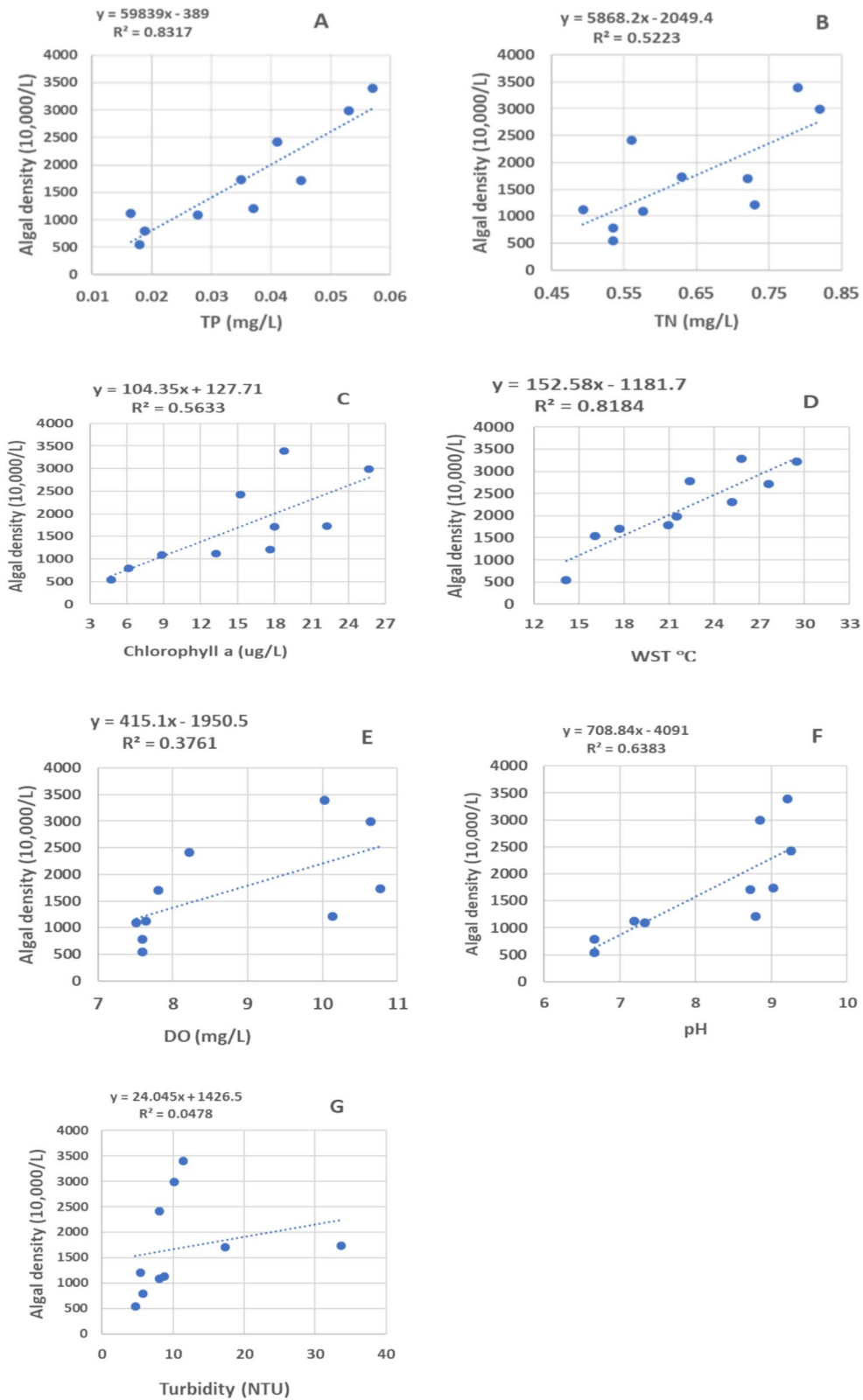
Low dissolved oxygen levels resulting from algal blooms can have adverse effects on aquatic ecosystems, potentially harming fish and other aquatic organisms that rely on oxygen for survival (Burdick et al., 2020). Monitoring both dissolved oxygen and algal bloom occurrence is crucial for understanding and managing water quality in reservoirs. The relationship between DO and algal density is shown in Fig. 11E. Changes in pH values and DO values are caused by physical disturbances (shipping and wind) and biochemical processes (respiration organic matter degradation) at the bottom of the water (Li et al., 2022).

For an accurate understanding of the effects of seasonality on phytoplankton dynamics, temperature is essential, especially when phytoplankton growth is considered (Fanesi et al., 2017; Grimaud et al., 2015). HABs are often dominated by specific species of algae that thrive under certain temperature conditions. Linear regressions revealed a positive relationship between surface water temperature and algae density (R^2 0.81). We found that higher temperatures accelerated the proliferation of HABs (Fig. 11D). Different species of algae have different temperature preference ranges for growth. For example, cyanobacteria, a common HAB species, tend to thrive in warm water with temperatures between 20 and 30 °C. Other types of algae may have different temperature preferences. In the Dongzhang Reservoir, higher cell density was recorded when the surface water temperature was

between 25 and 30 °C (Figs. 11 & 12). There was a greater density of cells in May, June, and July (2626 million/L, 3392.2 million/L, and 2418.4 million/L, respectively). The surface water temperature during this time ranged from 20 to 30 °C. A relatively high temperature, nutrient availability and low flow rate increase the vulnerability of a site to HAB events (Xin et al., 2020).

Higher temperatures prolong the duration of HABs; algal blooms can be triggered by increases in temperature, and once established, they can persist for extended periods in warm waters (Wells et al., 2015). This extended bloom duration can exacerbate the negative effects of HABs, such as oxygen depletion and toxin production (Yan et al., 2022). Understanding these complex interactions is critical for managing and mitigating the impacts of HABs on water resources (Summers & Ryder, 2023). Supplementary Fig. 1 shows the water surface temperature map of the Dongzhang Reservoir in 2022 in the inlet section of the Dongzhang Reservoir, which has higher temperatures than the other sections of the reservoir. We also discussed above the consistent variation trend of phytoplankton biomass in the inlet, center, and outlet sections of the reservoir, with the peak occurring in May. The inlet section had the highest phytoplankton biomass of the three sections studied (Fig. 6 & Table 2). In the inlet section, nutrient flow is also high (Chen, 2020), and a water surface map (supplementary Fig. S1) shows that the temperature is also high. During the summer, the temperature of the Dongzhang Reservoir is stable, which is suitable for algae growth (Chen, 2020; Zhen, 2017).

Temperature plays an important role in the development of HABs, but it does not act alone; other factors, such as nutrient availability, light availability, water flow, and grazing by zooplankton, may also interact with temperature to influence the growth and dynamics of HABs in aquatic reservoirs (Summers & Ryder, 2023). Our study also revealed that temperature is not the only proliferation factor. In 2023, the average water surface temperature was higher than that in 2022, but the algae density was greater in 2022, which means that another factor is also important for the proliferation of HABs. According to Chen (2020), cyanobacteria blooms in spring are derived mainly from sediment.



◀**Figure.11** Linear regression analysis between the algal density, TP (A), TN (B), chlorophyll-a (C), water surface temperature (D), DO (E), Ph (F) and turbidity (G) in the Dongzhang Reservoir based on 2022 data

HAB extraction from other band combinations

Furthermore, we processed Sentinel-2 images via a combination of different bands (NIR, SWIR1, and red-edge bands) to identify clusters of high relevance, integrate extraction findings across bands, and evaluate the severity of algal blooms. For mapping HABs in Dongzhang, we specifically selected the red-edge, NIR, and SWIR1 bands, which had not been previously utilized for HAB extraction in this reservoir. The combination of red-edge, NIR, and SWIR1 bands enhances HAB detection by providing complementary data on water and algal properties. Red-edge detects chlorophyll content, NIR distinguishes water bodies and algae, and SWIR1 identifies water turbidity and organic matter (Arias et al., 2025; Bramich et al., 2021a, 2021b; Khan et al., 2021; Wang et al., 2025; Yan et al., 2022). The presence of turbid water complicates HAB mapping via spectral bands within the visible range and shortwave infrared (SWIR2) bands, as they fail to distinguish between HABs and open water. HABs exhibit higher reflectances in the red-edge, NIR, and SWIR1 bands than turbid and open water does (Xu et al., 2021; Zhen, 2017). In the Erhai Lake watershed during autumn, significant correlations were observed between Sentinel-2 satellite data and chlorophyll-a mass concentrations, with (B6, B7/B6) and (B6+B8)/(B7+B8a) showing the highest correlation coefficients among single-band, single-band ratio, and dual-band ratio analyses (Junen, 2022). The results from the Sentinel-2 MSI images, including those extracted from bands 4, 5, 6, 7, 8, 8a, and 11, are presented in supplementary Figs. 2 & 3. Detailed spectral band information for Sentinel-2A is provided in supplementary Table 1.

Capacities of Sentinel-2 for the monitoring of HABs

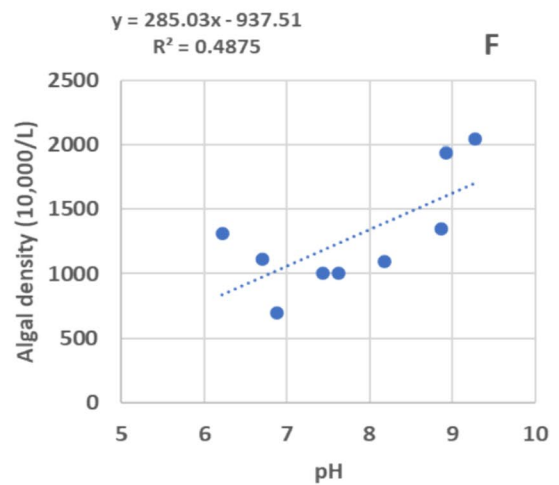
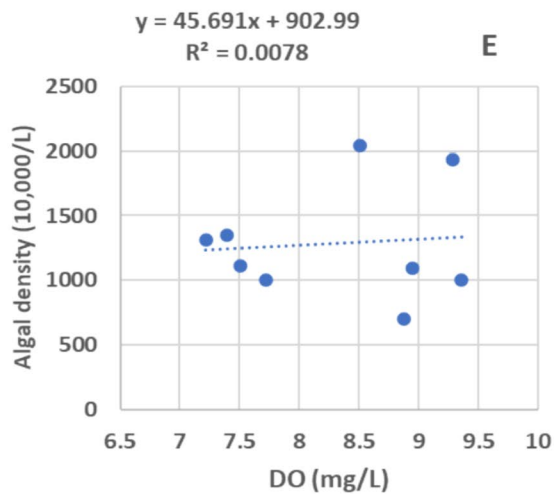
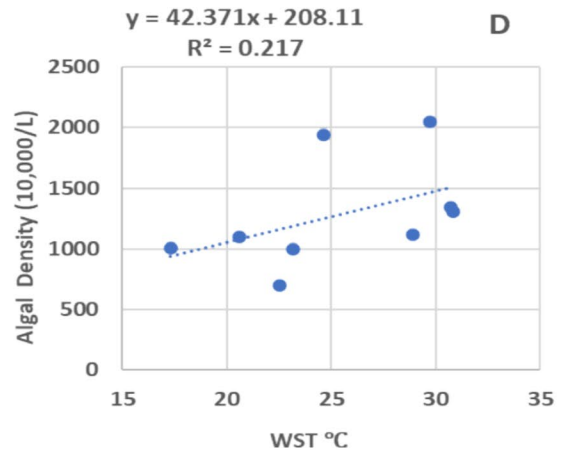
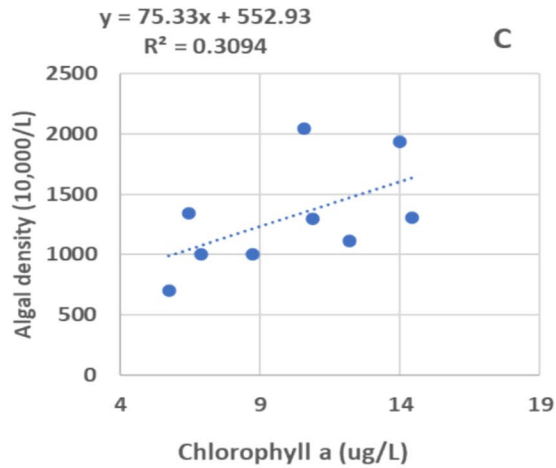
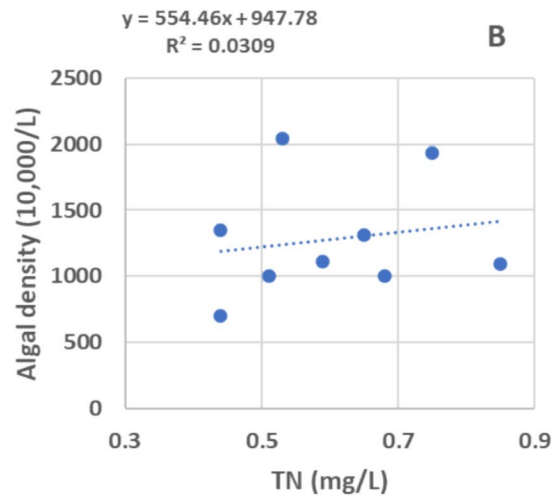
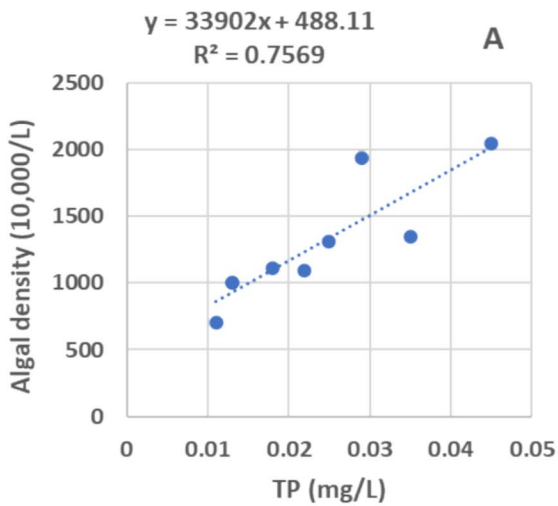
Various algorithms have been developed for remote sensing inversion of chlorophyll-a concentrations, each with limitations and sensor-specific adaptability (Yang et al., 2022). This study demonstrated the feasibility of identifying algal blooms in the Dongzhang Reservoir via Sentinel-2 imagery. Compared

with in situ sampling data, the NDCI provided the most accurate estimation of chlorophyll-a concentrations. Originally designed for coastal and marine environments, NDCI has proven effective in detecting chlorophyll-a in freshwater environments as well (Caballero & Navarro, 2021; Cillero Castro et al., 2020; Lobo et al., 2021; Weber et al., 2020). The high spatial and temporal resolution of Sentinel-2 has made it a popular choice for multispectral remote sensing in recent years, particularly in monitoring lake water environments and ecology (Lobo et al., 2021). The red-edge and NIR wavelengths, which are less affected by detritus and suspended particles typical in the Dongzhang Reservoir, yielded better NDCI and NDVI results across all the sampling locations (Zhen, 2017; Xu, 2022). Peak chlorophyll-a concentrations were observed in the Dongzhang Reservoir via the NDCI, which is consistent with previous findings (Mishra & Mishra, 2012).

Seasonal maps revealed the effectiveness of the NDCI in identifying blooms throughout the year, with the highest values recorded each month (Table 2). Red-edge wavelengths are more effective at detecting algal blooms than are NIR and narrow NIR wavelengths because of their reflectance characteristics (Caballero et al., 2020; Cillero Castro et al., 2020). Despite being designed primarily for land-based studies, Sentinel-2 sensors have shown promising capabilities in monitoring HABs (Caballero et al., 2020; Cao et al., 2021; Cillero Castro et al., 2020; Lobo et al., 2021). The performance of the NDVI in the Dongzhang Reservoir was less effective, likely because of the organic material and turbidity levels. Compared with the NIR-based indices, the blue-green indices may be more accurate when blooms are clearly visible on the water surface and penetrate slightly deeper (Legleiter et al., 2011). However, the NDCI was found to be more effective at capturing HABs than the NDVI was in this investigation.

Limitations of Sentinel-2 for the detection of HABs

This study has several limitations. First, comparing in situ and satellite observations introduces temporal mismatches (Table 2). Field sampling was conducted on 31 dates across three locations in the Dongzhang Reservoir between March and December 2022 (31 dates × 3 sites = 93 samples). Because of cloud cover and acquisition timing, some Sentinel-2 scenes were



◀**Fig. 12** Linear regression analysis between algal density, TP (A), TN (B), chlorophyll-a, (C), water surface temperature (D), DO (E), pH (F) and turbidity (G) in the Dongzhang Reservoir based on 2023 data

available only several days after sampling; approximately two-thirds of the scenes were acquired within 0–1 day of the field campaign (20/31), including 6/31 on the same day. A maximum gap of five days occurred in a subset of matchups, which may introduce uncertainty given the rapid onset and dissipation of blooms (Cillero Castro et al., 2020).

Second, the revisit and spatial resolution of Sentinel-2 can limit detection of fine-scale dynamics. Although 10–20 m pixels are suitable for many inland waters, small or highly patchy blooms may be under-resolved. Daily constellations (e.g., PlanetScope at 3 m; RapidEye at 5 m) can provide higher temporal resolution, but access, cost, and cloud cover remain practical constraints (Hu, 2021). Where feasible, fusing Sentinel-2 with Sentinel-3 OLCI and higher-resolution constellations (3–5 m) may better balance spatial and temporal coverage.

Third, Sentinel-2's multispectral bands do not reliably differentiate phytoplankton taxa and cannot indicate toxicity. As a result, species-level identification (e.g., toxin-producing cyanobacteria) and toxicity assessment require supporting lines of evidence (e.g., microscopy, pigment/protein assays, or targeted sensors). While Sentinel-3 OLCI includes a narrow band near 620 nm that enhances sensitivity to phycocyanin, its 300 m pixels are generally unsuitable for small water bodies. Hyperspectral approaches may improve species/guild detection, but current satellite options involve trade-offs among spatial, spectral, and temporal resolution (Caballero et al., 2020; Pahlevan et al., 2017).

Fourth, matchup uncertainties arise from measurement and processing choices. Reported differences between in situ and Sentinel-2 products in comparable studies underscore sensitivity to (i) in situ sampling depth and method, (ii) atmospheric correction and adjacency effects, (iii) algorithm parameterization and transferability across seasons and stations, and (iv) geolocation and mixed-pixel issues in shoreline or turbid environments. These factors can propagate into regression fits and error metrics.

Finally, our sampling design emphasized seasonal coverage but did not capture sub-daily variability.

Hourly to daily measurements (e.g., autonomous sensors or denser field campaigns) would better characterize short-lived bloom events and improve matchup statistics.

Conclusion

This work demonstrates that integrating Sentinel-2 imagery with in situ measurements can characterize spatiotemporal patterns of phytoplankton biomass and chlorophyll-a in the Dongzhang Reservoir. Across 2022, linear models yielded strong to moderate correlations between phytoplankton density and remote-sensing indices ($R^2=0.81$, 0.74, and 0.62). Dominant taxa, including Cyanobacteria (e.g., *Microcystis*), varied by season and station. Nutrient availability particularly TP and, to a lesser extent, TN was consistently associated with elevated bloom metrics. In 2022, TP and water temperature showed high explanatory power ($R^2=0.83$ and 0.81, respectively); in 2023, TP remained the primary predictor ($R^2=0.75$). Given these associations, management actions that reduce nutrient loads are likely to lower bloom risk. Examples include controlling agricultural runoff, improving wastewater treatment, and enforcing nutrient-discharge and land-use regulations. Operationally, closer monitoring of Cyanobacteria in spring and summer, increased attention to inlet reaches, and targeted nutrient controls are recommended. These findings should be interpreted with the limitations noted above, particularly temporal mismatches, spatial-resolution constraints, and species/toxicity ambiguity. Future work could quantify the roles of light and hydrodynamics using coupled hydrodynamic–biogeochemical models, higher-frequency in situ sensors, and, where feasible, multi-sensor satellite fusion or hyperspectral data.

Data availability statement

The datasets generated and analyzed during the current study are available from the corresponding author upon reasonable request.

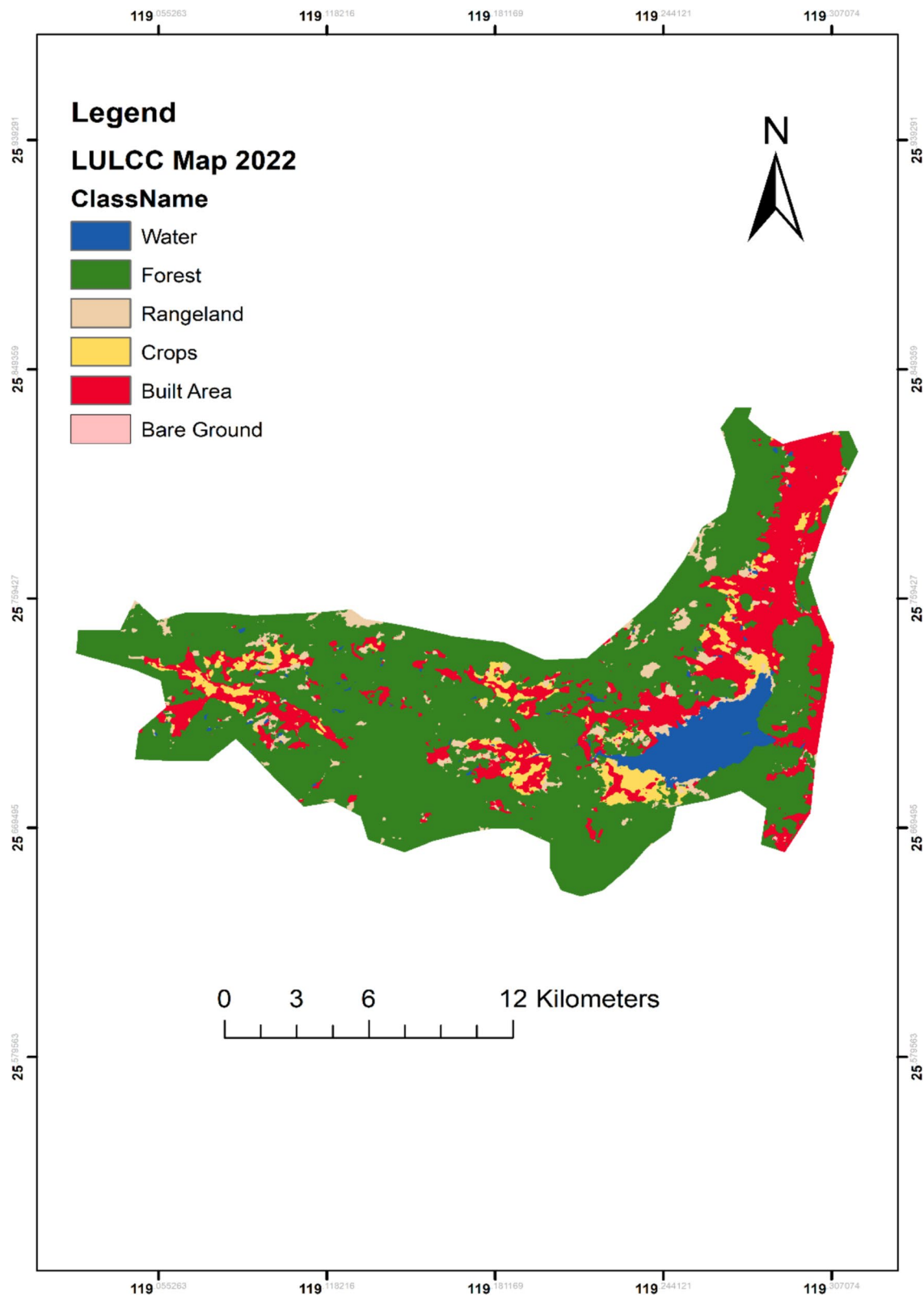


Fig. 13 LULCC analysis map of the Dongzhang reservoir catchments in 2022

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Declarations

Conflict of Interests All the authors have agreed to submit this manuscript to this journal. All the authors declare that they have no conflicts of interest.

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